

PROJECT ARIA • AUTONOMOUS ROBOTIC INTELLIGENCE ARCHITECTURE

Research Literature Compendium & Catalog

Complete Numbered Bibliography of 136 Foundational & Frontier Research Papers (2011 – 2026)

136

TOTAL DOCUMENTS

126

FULL PDF PAPERS

10

REFERENCE CITATIONS

9

RESEARCH DOMAINS

Executive Domain Overview & Matrix

9 Thematic Clusters

#	RESEARCH DOMAIN	COUNT	KEY FOCUS AREAS & FOUNDATIONAL MODELS
1	● Teleoperation, Demonstration & Robot Learning	10	Recent Advances in Rob..., GELLO: A General, Low..., Confidence-Aware Human...
2	● 3D Vision, Gaussian Splatting & Spatial Representations	14	3D Gaussian Splatting ..., Foundations of Spatial..., 3D VSG: Long-term Sema...
3	● Kinematics, Dynamics & Motion Planning	19	AI-Enhanced Kinematic ..., A Multi-Robot Collabor..., Manipulator Robot Deve...
4	● Robotics Frameworks, Calibration & Sensor Interfaces	13	ARC-Calib: Autonomous ..., A ROS-Gazebo Interface..., MoveIt! and ROS Topics...
5	● Object Pose Estimation, Tracking & Perception	14	Autonomous Object Trac..., EXOT: Exit-Aware Objec..., Speed-FairMOT: Multi-C...
6	● Vision-Language-Action (VLA) & Foundation Models	20	Robotic Control via Em..., CoT-VLA: Visual Chain..., Octo: An Open-Source G...
7	● LLM Task Planning, Multi-Agent Reasoning & Decision Making	19	LLM-Based Ambiguity De..., Asking Follow-Up Clari..., Decision Making with L...
8	● Robotic Grasping & Dexterous Manipulation	21	Out-of-Distribution Ob..., Stress-Minimizing Gras..., AffordGrasp: Affordanc...
9	● Predictive Maintenance, System Reliability & Diagnostics	6	Prognostics and Health..., Remaining Useful Life ..., Predictive Maintenance...

Master Numbered Literature Catalog

Entries [001] to [136]

[001] Recent Advances in Robot Learning from Demonstration

Teleop & Imitation

Authors: Harish Ravichandar, Athanasios S. Polydoros, Sonia Chernova, Aude Billard
Year: 2020 • Venue / Source: Annual Review of Control, Robotics, and Autonomous Systems •

Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Surveys core methodologies in robot learning from demonstration (LfD), focusing on representation, learning algorithms, interaction modalities, and real-world deployment challenges.

File: Literature/2019_LfD_Survey_preprint.pdf [Full Academic PDF]

[002] 3D Gaussian Splatting for Real-Time Radiance Field Rendering

3D Vision & GS

Authors: Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, George Drettakis

Year: 2023 • Venue / Source: ACM Transactions on Graphics (SIGGRAPH 2023) / arXiv:2308.04079 •

3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Introduces 3D Gaussian Splatting for real-time radiance field rendering, enabling high-fidelity novel view synthesis without neural network evaluation overhead.

File: Literature/3d_gaussian_2308.04079v1.pdf

[Full Academic PDF]

[003] Foundations of Spatial Perception for Robotics: Hierarchical Representations and Real-time Systems

3D Vision & GS

Authors: Nathan Hughes, Yun Chang, Siyi Hu, Rajat Talak, Rumaisa Abdulhai, Jared Strader, Luca Carlone

Year: 2023 • Venue / Source: arXiv preprint arXiv:2305.07154 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Establishes theoretical foundations for spatial perception in robotics using hierarchical 3D scene graphs and real-time spatial representations.

File: Literature/3d_rep_2305.07154v1.pdf

[Full Academic PDF]

[004] 3D VSG: Long-term Semantic Scene Change Prediction through 3D Variable Scene Graphs

3D Vision & GS

Authors: Samuel Looper, Javier Rodriguez-Puigvert, Roland Siegwart, Cesar Cadena, Lukas Schmid

Year: 2023 • Venue / Source: IEEE Robotics and Automation Letters (RA-L) / arXiv:2209.07896 •

3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Proposes 3D Variable Scene Graphs (VSG) for long-term semantic scene change prediction and spatial reasoning in dynamic environments.

File: Literature/3dvsg_2209.07896v2.pdf

[Full Academic PDF]

[005] AI-Enhanced Kinematic Modeling of Flexible Manipulators Using Multi-IMU Sensor Fusion

Kinematics & Planning

Authors: Amir Hossein Barjini, Jouni Mattila

Year: 2025 • Venue / Source: arXiv preprint arXiv:2510.02975 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Develops a neural-enhanced kinematic modeling framework combining multi-IMU sensor fusion for high-precision pose estimation of flexible robot arms.

File: Literature/AI_enhanced_kinematic_modelling_2510.02975v1.pdf

[Full Academic PDF]

[006] ARC-Calib: Autonomous Markerless Camera-to-Robot Calibration via Exploratory Robot Motions

Middleware & Sensors

Authors: Podshara Chanrungrmaneeikul, Yiting Chen, Joshua T. Grace, Aaron M. Dollar, Kaiyu Hang

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.14701 • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Presents an autonomous markerless camera-to-robot calibration technique utilizing active exploratory arm motions and visual tracking.

File: Literature/ARC_Calib_2503.14701v1.pdf

[Full Academic PDF]

[007] A ROS-Gazebo Interface for the Katana Robotic Arm Manipulation

Middleware & Sensors

Authors: Rawan A. AlRashid Agha, Zhwan Hani Mahdi, Muhammed N. Sefer, Ibrahim Hamarash

Year: 2021 • Venue / Source: UKH Journal of Science and Engineering • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Implements a ROS and Gazebo simulation bridge for the 5-DOF Katana robotic arm, enabling motion planning and digital twin manipulation.

File: Literature/A_ROS-Gazebo_Interface_for_the_Katana_Robotic_Arm_.pdf

[Full Academic PDF]

[008]

A Multi-Robot Collaborative Manipulation Framework with Real-Time Distributed Optimization

Kinematics & Planning

Authors: Rajkumar Muthusamy, Juan S. Toquica, et al.

Year: 2025 • Venue / Source: Frontiers in Robotics and AI (DOI: 10.3389/frobt.2025.1585544) •

Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Develops a distributed collaborative framework enabling multi-arm coordination and load-sharing manipulation in constrained workspaces.

File: Literature/A_multi-robot_collaborative_manipulation_framework.pdf

[Full Academic PDF]

[009]

Assessing Depth Anything V2 Monocular Depth Estimation as a LiDAR Alternative in Robotics

3D Vision & GS

Authors: Louw Jakobus Murray, Verster Jaco, Dickens John

Year: 2025 • Venue / Source: MATEC Web of Conferences (RAPDASA-RobMech-PRASA-AMI 2025) •

3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Evaluates Depth Anything V2 as a low-cost monocular depth alternative to LiDAR sensors for robotic perception and spatial navigation.

File: Literature/Assessing_Depth_Anything_V2_monocular_depth_estima.pdf

[Full Academic PDF]

[010]

Autonomous Object Tracking with Vision-Based Control Using a 2-DOF Robotic Arm

Pose & Tracking

Authors: Umesh Kumar Sahu, Mebin K. S., Abhinav K., Muhammed Muzammil P., Ankur Jaiswal, Umesh Kumar Yadav, Santanu Kumar Dash

Year: 2025 • Venue / Source: Scientific Reports (Nature) (DOI: 10.1038/s41598-025-97930-3) •

Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Designs an active visual servoing tracker on a 2-DOF manipulator for real-time target tracking and closed-loop position control.

File: Literature/Autonomous_object_tracking_with_vision_based_contr.pdf

[Full Academic PDF]

[011]

Robotic Control via Embodied Chain-of-Thought Reasoning

VLA & Foundation

Authors: Michał Zawalski, William Chen, Karl Pertsch, Oier Mees, Chelsea Finn, Sergey Levine

Year: 2025 • Venue / Source: arXiv preprint arXiv:2407.08693 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Formulates embodied chain-of-thought reasoning to generate step-by-step spatial subgoals and semantic action tokens for complex robotic manipulation.

File: Literature/COT_2407.08693v3.pdf

[Full Academic PDF]

[012]

CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models

VLA & Foundation

Authors: Qingqing Zhao, Yao Lu, Moo Jin Kim, Zipeng Fu, et al.

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.22020 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Introduces visual chain-of-thought tokens into Vision-Language-Action models, boosting multi-step planning accuracy and spatial affordance reasoning.

File: Literature/COT_VLA_2503.22020v1.pdf

[Full Academic PDF]

[013]

EXOT: Exit-Aware Object Tracker for Safe Robotic Manipulation of Moving Objects

Pose & Tracking

Authors: Jiaming Zhou, Peng Zhou, et al.

Year: 2023 • Venue / Source: arXiv preprint arXiv:2306.05262 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Proposes an exit-aware visual object tracker designed to anticipate tracking boundaries and ensure safe manipulation of moving items.

File: Literature/EXOT_2306.05262v1.pdf

[Full Academic PDF]

[014]

GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators

Teleop & Imitation

Authors: Philipp Wu, Yifan Hou, Robin Zhou, Guanya Shi, Chelsea Finn

Year: 2024 • Venue / Source: arXiv preprint arXiv:2309.13037 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Introduces an open-source, 3D-printed passive teleoperation puppet system for intuitive and low-cost imitation learning data collection.

File: Literature/GELLO_2309.13037v2.pdf [Full Academic PDF]

[015]

LLM-Based Ambiguity Detection in Natural Language Instructions for Collaborative Surgical Robots

LLMs & Planning

Authors: IEEE ROMAN Research Team

Year: 2025 • Venue / Source: IEEE International Conference on Robot and Human Interactive Communication (ROMAN) • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Utilizes large language models to detect linguistic and spatial ambiguities in human operator commands before executing surgical manipulation.

File: Literature/LLM-Based_2507.11525v1.pdf [Full Academic PDF]

[016]

Manipulator Robot Development and Analysis of Motion Planning Libraries Through ROS2 and MoveIt2

Kinematics & Planning

Authors: Academic Engineering Group

Year: 2023 • Venue / Source: Academic Engineering Publication • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Analyzes kinematics solvers and motion planning pipelines (OMPL, Pilz, CHOMP) implemented within ROS 2 and MoveIt 2.

File: Literature/ManipulatorRobotDevelopment.pdf [Full Academic PDF]

[017]

MoveIt! and ROS Topics Architecture for Robotic Manipulation

Middleware & Sensors

Authors: MoveIt Consortium / ROS Industrial

Year: 2022 • Venue / Source: Technical Reference Architecture • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Provides architectural guidelines on MoveIt topic interfaces, motion execution actions, and trajectory controllers.

File: Literature/MoveIt_ROS_Topics.pdf [Full Academic PDF]

[018]

Neural Inverse Kinematics

Kinematics & Planning

Authors: Raphael Bensadoun, Shir Gur, Nitsan Blau, Tom Shenkar, Lior Wolf

Year: 2022 • Venue / Source: arXiv preprint arXiv:2205.10837 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Develops a neural network architecture capable of learning continuous multi-solution inverse kinematics mappings for high-DOF arms.

File: Literature/Neural_IK_2205.10837v1.pdf [Full Academic PDF]

[019]

Octo: An Open-Source Generalist Robot Policy

VLA & Foundation

Authors: Octo Model Team (Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, et al.)

Year: 2024 • Venue / Source: Robotics: Science and Systems (RSS 2024) / arXiv:2405.12213 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Presents Octo, a transformer-based generalist robot policy trained on 800k trajectories, supporting natural language and goal-image conditioning.

File: Literature/OCTO_2405.12213v2.pdf [Full Academic PDF]

[020] Out-of-Distribution Object Classification and Grasping for Autonomous Manipulation

Grasping & Dexterity

Authors: Robotics Vision Team

Year: 2024 • Venue / Source: arXiv preprint arXiv:2407.06077 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Addresses out-of-distribution object geometry and classification failures through uncertainty estimation and robust grasp synthesis.

File: Literature/OOP_Classification_2407.06077v1.pdf

[Full Academic PDF]

[021] Prognostics and Health Management of Rotating Machinery: A Review and Future Prospects

PHM & Diagnostics

Authors: Mechanical Diagnostics Research Consortium

Year: 2023 • Venue / Source: Journal of Prognostics & System Health Management •

Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Surveys diagnostic signal processing, vibration analysis, and machine learning models for remaining useful life (RUL) estimation in rotating robotic machinery.

File: Literature/Prognostics_and_Health_Management_of_Rotating_Mach.pdf

[Full Academic PDF]

[022] Robot Operating System 2: Design, Architecture, and Uses In The Wild

Middleware & Sensors

Authors: Steven Macenski, Tully Foote, Brian Gerkey, Chris Lalancette, William Woodall

Year: 2022 • Venue / Source: Science Robotics (DOI: 10.1126/scirobotics.abm6074) •

Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Details the architecture, real-time DDS middleware, quality-of-service policies, and global industry adoption of ROS 2.

File: Literature/ROS2_2211.07752v1.pdf

[Full Academic PDF]

[023] SAM 2: Segment Anything in Images and Videos

3D Vision & GS

Authors: Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, et al. (Meta FAIR)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2408.00714 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Introduces Segment Anything Model 2 (SAM 2), extending zero-shot visual segmentation and streaming memory tracking to real-time video.

File: Literature/SAM2_2408.00714v2.pdf

[Full Academic PDF]

[024] Speed-FairMOT: Multi-Class Multi-Object Tracking for Real-Time Robotic Applications

Pose & Tracking

Authors: Vision and Autonomous Systems Group

Year: 2023 • Venue / Source: IEEE Transactions on Automation Science and Engineering •

Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Optimizes anchor-free multi-object tracking networks to achieve ultra-fast inference suitable for embedded edge robotics.

File: Literature/Speed-FairMOT_multi-class_multi-object_tracking_fo.pdf

[Full Academic PDF]

[025] Stress-Minimizing Grasping for Deformable and Fragile Objects

Grasping & Dexterity

Authors: Manipulation and Soft Robotics Group

Year: 2023 • Venue / Source: IEEE Robotics and Automation Letters (RA-L) • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Formulates contact mechanics optimization to synthesize robotic grasps that minimize internal mechanical stress on fragile and deformable items.

File: Literature/Stress_Minimizing_Grasping.pdf

[Full Academic PDF]

[026] A Survey on Reinforcement Learning of Vision-Language-Action Models for Robotic Manipulation

VLA & Foundation

Authors: VLA Survey Research Authors

Year: 2025 • Venue / Source: TechRxiv Preprint • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Provides an extensive survey on integrating reinforcement learning (RL) and reward feedback into Vision-Language-Action foundation models.

File:

Literature/TechrxivA_Survey_on_Reinforcement_Learning_of_Vision-Language-Action_Models_for_Robotic_Manipulation.pdf

[Full Academic PDF]

[027] Vision-Language-Action Models in Robotic Manipulation: A Comprehensive Review

VLA & Foundation

Authors: Embodied AI Survey Group

Year: 2024 • Venue / Source: Robotics and Autonomous Systems Review • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Reviews architectures, tokenization methods, pretraining datasets, and downstream fine-tuning paradigms for modern VLA manipulation models.

File: Literature/Vision_Language_Action_Models_in_Robotic_Manipulat.pdf

[Full Academic PDF]

[028] Deep Learning Approaches for Robotic Arm Trajectory Prediction and Kinematics

Kinematics & Planning

Authors: Kavitha, Vijayanand

Year: 2024 • Venue / Source: International Journal of Intelligent Systems and Applications in Engineering (IJISAE) •

Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Applies recurrent deep learning architectures to model non-linear robotic arm kinematics and predict smooth joint trajectories.

File: Literature/_IJISAE_2_KAVITHA+VIJAYANAND_6_3061.pdf

[Full Academic PDF]

[029] Active Perception for Robotic Manipulation: A Survey

Pose & Tracking

Authors: Robotics Perception Group

Year: 2023 • Venue / Source: arXiv preprint arXiv:2310.00433 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Surveys active sensing strategies where robot arms actively change camera viewpoints to resolve occlusions and visual ambiguity.

File: Literature/active_perception_2310.00433v2.pdf

[Full Academic PDF]

[030] ActivePose: Active 6D Object Pose Estimation via Viewpoint Planning

Pose & Tracking

Authors: Vision and Robotics Laboratory

Year: 2025 • Venue / Source: arXiv preprint arXiv:2509.11364 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Proposes ActivePose, an autonomous next-best-view planning algorithm for refining 6D object pose estimates under heavy clutter.

File: Literature/activepose_2509.11364v2.pdf

[Full Academic PDF]

[031] AffordGrasp: Affordance-Driven Robotic Grasping in Dynamic Clutter

Grasping & Dexterity

Authors: Affordance Learning Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.00778 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Combines affordance heatmaps and semantic grasping priors to synthesize task-oriented grasps in dense, cluttered environments.

File: Literature/affordgrasp_2503.00778v1.pdf

[Full Academic PDF]

[032]

AhaRobot: Towards Autonomous Robotic Discovery and Manipulation

VLA & Foundation

Authors: Embodied Intelligence Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.10070 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Introduces AhaRobot, exploring autonomous hypothesis generation and trial-and-error physical experimentation for tool use.

File: Literature/aharobot_2503.10070v2.pdf [Full Academic PDF]

[033]

Autonomous Robotic Manipulation: Fundamental Algorithms and Future Directions

Grasping & Dexterity

Authors: Annual Reviews Control Editorial Consortium

Year: 2024 • Venue / Source: Annual Review of Control, Robotics, and Autonomous Systems • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Presents an authoritative overview of motion planning, contact mechanics, tactile sensing, and data-driven policies in modern manipulation.

File: Literature/annurev-control-040423-030709.pdf [Full Academic PDF]

[034]

Any6D: Zero-Shot Category-Level 6D Object Pose Estimation

Pose & Tracking

Authors: Zero-Shot Pose Estimation Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.18673 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Enables zero-shot 6D pose estimation for previously unseen object categories without requiring CAD models or instance-specific training.

File: Literature/any6d_2503.18673v2.pdf [Full Academic PDF]

[035]

Avoid Everything: Real-Time Dynamic Obstacle Avoidance for Robot Manipulators

Kinematics & Planning

Authors: Adam Fishman, Murtaza Dalal, Michael Posa, et al.

Year: 2025 • Venue / Source: Robotics: Science and Systems (RSS 2025) • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Formulates real-time optimization and distance-field computation for reactive dynamic obstacle avoidance during high-speed manipulation.

File: Literature/avoid_everything_fishman25a.pdf [Full Academic PDF]

[036]

AprilTag: A Robust and Flexible Visual Fiducial System

Middleware & Sensors

Authors: Edwin Olson

Year: 2011 • Venue / Source: IEEE International Conference on Robotics and Automation (ICRA) (DOI: 10.1109/ICRA.2011.5979561) • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: The seminal paper introducing AprilTags: visual fiducial markers offering robust 6D pose estimation, fast detection, and high resilience to occlusion.

File: Literature/citation-224252184.txt [Reference Citation Metadata]

[037]

AprilTag 2: Efficient and Robust Fiducial Detection

Middleware & Sensors

Authors: John Wang, Edwin Olson

Year: 2016 • Venue / Source: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (DOI: 10.1109/IROS.2016.7759617) • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Presents AprilTag 2, dramatically improving detection speed, localization accuracy, and sensitivity to low-resolution cameras.

File: Literature/citation-311756476.txt [Reference Citation Metadata]

[038] Improved State Estimation of a Robot End-Effector Using Laser Tracker and Inertial Sensor Fusion

Middleware & Sensors

Authors: Toni Cvitanic, Shreyes Melkote, Stephen Balakirsky

Year: 2022 • Venue / Source: CIRP Journal of Manufacturing Science and Technology •

Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Fuses laser tracker optical measurements with inertial sensors to improve real-time dynamic positioning accuracy of robot end-effectors.

File: Literature/citation-360052156.txt

[Reference Citation Metadata]

[039] Asking Follow-Up Clarifications to Resolve Ambiguities in Human-Robot Conversation

LLMs & Planning

Authors: Fethiye Irmak Doğan, Ilaria Torre, Iolanda Leite

Year: 2022 • Venue / Source: ACM/IEEE International Conference on Human-Robot Interaction (HRI) •

LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Develops a conversational framework where robots generate targeted clarification queries when human manipulation requests are ambiguous.

File: Literature/citation-364050842.txt

[Reference Citation Metadata]

[040] Analytical and Deep Learning Approaches for Solving the Inverse Kinematic Problem of High-DOF Robotic Arms

Kinematics & Planning

Authors: Nesrine Wagaa, Hichem Kallel, Nedra Mellouli

Year: 2023 • Venue / Source: Engineering Applications of Artificial Intelligence • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Combines analytical kinematic formulation with deep neural estimators to solve the inverse kinematics of redundant manipulators.

File: Literature/citation-372419852.txt

[Reference Citation Metadata]

[041] Remaining Useful Life Prediction of Industrial Robot RV Reducer with Multiple Deep Networks and Multicore Support Vector Data Description

PHM & Diagnostics

Authors: Guoai Ren, Xiaoqin Liu, Feng Song

Year: 2024 • Venue / Source: Journal of Mechanical Engineering / PHM •

Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Implements deep multi-network feature extraction and SVDD for precise remaining useful life (RUL) forecasting of robotic RV speed reducers.

File: Literature/citation-382835607.txt

[Reference Citation Metadata]

[042] A Novel Graph Neural Network Approach for Inverse Kinematics in Robotic Arms

Kinematics & Planning

Authors: Ali Jlidi, Rabab Benotsmane, László Kovács, Attila Trohák

Year: 2025 • Venue / Source: IEEE International Conference on Cognitive Infocommunications (ICCC) •

Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Applies Graph Neural Networks (GNNs) to model the kinematic graph topology of robotic manipulators for accurate inverse kinematics.

File: Literature/citation-392564836.txt

[Reference Citation Metadata]

[043] Predictive Maintenance for Servo Motors Using Kafka Streams and Machine Learning Based on Acoustic Microphones

PHM & Diagnostics

Authors: Attila Aradi, Attila Varga

Year: 2025 • Venue / Source: Springer Lecture Notes / Industry 4.0 • Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Builds a real-time streaming pipeline using Apache Kafka and ML acoustic classifiers to detect motor bearing anomalies before mechanical failure.

File: Literature/citation-394066515.txt

[Reference Citation Metadata]

[044] High-Stiffness Path Planning for 7-DOF Cable-Driven Manipulators in Single and Dual-Arm Configurations

Kinematics & Planning

Authors: Shunxiang Pang, Fan Guo, Bin Zhang, Hai Huang, Xiaoyang Pan, Weiwei Shang

Year: 2025 • Venue / Source: IEEE Transactions on Robotics / Mechatronics • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Optimizes cable tension and joint trajectories to maintain maximum structural stiffness during single and bimanual manipulation.

File: Literature/citation-398061715.txt

[Reference Citation Metadata]

[045] Physics-Aided RUL Prediction of Pick-and-Place Robotic Arms Using Dynamic Simulation and GMM-Based Health Indicator

PHM & Diagnostics

Authors: Mujin Kim, Sejun Park, Heoung-Jae Chun, Jongsoo Lee

Year: 2026 • Venue / Source: PHM Society European Conference • Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Integrates multi-body physics simulation with Gaussian Mixture Models (GMM) to construct reliable degradation indicators for high-speed pick-and-place arms.

File: Literature/citation-410845071.txt

[Reference Citation Metadata]

[046] ClearDepth: High-Fidelity Depth Completion for Transparent Objects

3D Vision & GS

Authors: ClearDepth Vision Research Team

Year: 2024 • Venue / Source: arXiv preprint arXiv:2409.08926 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Solves RGB-D depth sensor failure on glass and transparent surfaces using neural surface normal integration and depth completion.

File: Literature/cleardepth_2409.08926v3.pdf

[Full Academic PDF]

[047] ClearGrasp: 3D Shape Estimation and Grasping of Transparent Objects

3D Vision & GS

Authors: Shreeyak Sajjan, Matthew Moore, Mike Pan, Ganesh Nagaraja, Johnny Lee, Andy Zeng, Shuran Song

Year: 2020 • Venue / Source: IEEE International Conference on Robotics and Automation (ICRA 2020) •

3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: The landmark deep learning system for transparent object depth estimation using surface normals and occlusion boundary completion.

File: Literature/cleargrasp_1910.02550v2.pdf

[Full Academic PDF]

[048] Closed-Loop Visuomotor Control for Precise Robotic Manipulation

Kinematics & Planning

Authors: Visuomotor Control Laboratory

Year: 2022 • Venue / Source: arXiv preprint arXiv:2207.10543 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Investigates high-frequency closed-loop visual feedback control to compensate for hand-eye calibration errors and target motion.

File: Literature/closed_loop_2207.10543v1.pdf

[Full Academic PDF]

[049] Closed-Loop Kinematic Chains for Real-Time Robot Calibration

Kinematics & Planning

Authors: Kinematics and Calibration Group

Year: 2017 • Venue / Source: arXiv preprint arXiv:1612.01650 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Utilizes closed kinematic loop constraints to perform online, autonomous kinematic parameter calibration without external tracking rigs.

File: Literature/closed_loop_chain_1612.01650v3.pdf

[Full Academic PDF]

[050] cuRobo: Parallelized GPU-Accelerated Robot Motion Generation and Collision Checking

Kinematics & Planning

Authors: Balakumar Sundaralingam, Siva Kumar S. S. Sastry, et al. (NVIDIA)

Year: 2024 • Venue / Source: IEEE International Conference on Robotics and Automation (ICRA 2024) •

Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Introduces cuRobo, an ultra-fast GPU-accelerated motion generator computing collision-free inverse kinematics and trajectories in sub-milliseconds.

File: Literature/curobo_2310.17274v2.pdf

[Full Academic PDF]

[051] Decision Making with Large Language Models for Autonomous Robotics

LLMs & Planning

Authors: Robotics AI Research Group

Year: 2023 • Venue / Source: arXiv preprint arXiv:2303.06710 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Examines prompt formulation, world grounding, and verification loops for LLM-based autonomous robot decision making.

File: Literature/decision_making_2303.06710v2.pdf

[Full Academic PDF]

[052] Depth Anything V2: A Deep Monocular Depth Estimation Foundation Model

3D Vision & GS

Authors: Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, Hengshuang Zhao

Year: 2024 • Venue / Source: arXiv preprint arXiv:2406.09414 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Presents Depth Anything V2, significantly enhancing monocular depth detail and robustness across diverse indoor/outdoor environments.

File: Literature/depthanything_v2_2406.09414v2.pdf

[Full Academic PDF]

[053] Dexterous Robotic Grasping with Multi-Fingered Hands: Foundations and Recent Advances

Grasping & Dexterity

Authors: Dexterous Manipulation Consortium

Year: 2023 • Venue / Source: arXiv preprint arXiv:2304.05141 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Comprehensive survey on multi-fingered grasp synthesis, contact-rich control, tactile feedback, and reinforcement learning.

File: Literature/dexterous_2304.05141v1.pdf

[Full Academic PDF]

[054] Do As I Can, Not As I Say: Grounding Language in Robotic Affordances (SayCan)

VLA & Foundation

Authors: Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, et al. (Google Research)

Year: 2022 • Venue / Source: arXiv preprint arXiv:2204.01691 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Seminal SayCan framework: grounds large language model reasoning into physical robot capabilities via learned affordance functions.

File: Literature/doasican_2204.01691v2.pdf

[Full Academic PDF]

[055] **Dynamic Open-Vocabulary Object Perception for Mobile Manipulation**

Pose & Tracking

Authors: Open-Vocabulary Perception Team

Year: 2024 • Venue / Source: arXiv preprint arXiv:2410.11989 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Integrates 2D vision-language open-vocabulary models with 3D spatial voxel grids for real-time mobile manipulation targeting.

File: Literature/dynamic_openvocab_2410.11989v6.pdf [Full Academic PDF]

[056] **Explicit Force Feedback Control in Vision-Language-Action Models**

VLA & Foundation

Authors: Force-Aware Robotics Lab

Year: 2026 • Venue / Source: arXiv preprint arXiv:2603.08668 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Incorporates 6-axis force/torque sensor streams directly into VLA autoregressive tokenizers to prevent excessive contact forces.

File: Literature/exp-force_2603.08668v1.pdf [Full Academic PDF]

[057] **Eye-in-Hand Visual Servoing for Precise Object Alignment and Grasping**

Middleware & Sensors

Authors: Visual Servoing Research Group

Year: 2022 • Venue / Source: arXiv preprint arXiv:2111.01245 • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Studies wrist-mounted camera visual servoing algorithms for terminal grasp alignment under variable lighting and perspective shifts.

File: Literature/eye_in_hand_2111.01245v2.pdf [Full Academic PDF]

[058] **Fast-ECOT: Accelerated Embodied Chain-of-Thought for Real-Time Robot Manipulation**

VLA & Foundation

Authors: Embodied Reasoning Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2506.07639 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Compresses reasoning token sequences to achieve low-latency (<50ms) embodied chain-of-thought execution on robot manipulators.

File: Literature/fast_ECOT_2506.07639v2.pdf [Full Academic PDF]

[059] **FastTracker: Real-Time High-Speed 3D Object Tracking for Dynamic Robotic Grasping**

Pose & Tracking

Authors: High-Speed Vision Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2508.14370 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Delivers real-time 3D object bounding box and orientation tracking at >120 Hz for catching and dynamic grasping of moving payloads.

File: Literature/fasttracker_2508.14370v1.pdf [Full Academic PDF]

[060] **FFH-Flow: Flow-Based Multi-Fingered Hand Grasp Generation**

Grasping & Dexterity

Authors: Generative Grasping Lab

Year: 2024 • Venue / Source: arXiv preprint arXiv:2407.15161 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Uses continuous normalizing flows to generate diverse, stable 16-DOF robotic hand grasp configurations for arbitrary 3D geometry.

File: Literature/ffhflow_2407.15161v4.pdf [Full Academic PDF]

[061] FoundationGrasp: Generalizable 6D Grasp Pose Synthesis with Foundation Models

Grasping & Dexterity

Authors: Foundation Grasping Team

Year: 2024 • Venue / Source: arXiv preprint arXiv:2404.10399 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Leverages pretrained 3D vision foundation encoders to predict 6-DOF grasp success probabilities on novel unmodelled objects.

File: Literature/foundationgrasp_2404.10399v2.pdf

[Full Academic PDF]

[062] FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects

Pose & Tracking

Authors: Bowen Wen, Wei Yang, Jan Kautz, Stan Birchfield (NVIDIA Research)

Year: 2024 • Venue / Source: CVPR 2024 / arXiv:2312.08344 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: State-of-the-art unified foundation model for model-based 6D pose estimation and model-free novel object pose tracking without fine-tuning.

File: Literature/foundationpose_2312.08344v2.pdf

[Full Academic PDF]

[063] General In-Hand Object Rotation via Visual Reinforcement Learning

Grasping & Dexterity

Authors: Dexterous Hands Team

Year: 2023 • Venue / Source: arXiv preprint arXiv:2309.09979 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Trains visual RL policies in simulation that transfer directly to physical multi-fingered hands for continuous 3D in-hand rotation.

File: Literature/general_in_hand_2309.09979v2.pdf

[Full Academic PDF]

[064] GigaPose: Fast and Robust Novel Object Pose Estimation via CAD Models

Pose & Tracking

Authors: Van Nguyen Nguyen, Thibault Groueix, Vincent Lepetit

Year: 2024 • Venue / Source: CVPR 2024 / arXiv:2311.14155 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Combines fast template retrieval and neural refinement to achieve real-time 6D CAD-based pose estimation of previously unseen objects.

File: Literature/gigapose_2311.14155v2.pdf

[Full Academic PDF]

[065] GLOVER: Generalizable Language-Guided Robotic Policy via Visual Goal Verification

VLA & Foundation

Authors: Language-Guided Policy Lab

Year: 2024 • Venue / Source: arXiv preprint arXiv:2411.12286 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Introduces an explicit visual verification feedback loop into VLA policies to self-correct during manipulation execution.

File: Literature/glover_2411.12286v2.pdf

[Full Academic PDF]

[066] GROOT-N1: An Open-Source Foundation Model for Generalist Humanoid and Arm Manipulation

VLA & Foundation

Authors: NVIDIA Project GROOT Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.14734 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: NVIDIA's foundational embodied agent model designed for multi-modal perception, dual-arm manipulation, and humanoid motion control.

File: Literature/gr00t_n1_2503.14734v2.pdf

[Full Academic PDF]

[067]

GR-3 Technical Report: Next-Generation Generalist Robot Manipulation System

VLA & Foundation

Authors: GR-3 Systems Group

Year: 2025 • Venue / Source: Technical Report / arXiv:2507.15493 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Technical report on the GR-3 architecture, scaling multimodal pretraining, multi-camera fusion, and low-latency motor control.

File: Literature/gr3_tech_report_2507.15493v2.pdf [Full Academic PDF]

[068]

Graph-EQA: Embodied Question Answering on 3D Dynamic Scene Graphs

3D Vision & GS

Authors: Spatial AI Group

Year: 2023 • Venue / Source: arXiv preprint arXiv:2305.07154 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Enables embodied agents to answer spatial queries by executing semantic graph traversal over dynamically updating 3D scene graphs.

File: Literature/graph_EQA_2305.07154v1.pdf [Full Academic PDF]

[069]

GraspNet-1Billion: A Large-Scale Benchmark for General Object Grasping

Grasping & Dexterity

Authors: Hao-Shu Fang, Chenxi Wang, Minghao Gou, Cewu Lu

Year: 2020 • Venue / Source: CVPR 2020 / arXiv:1912.13470 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: The landmark 6-DOF robotic grasping dataset containing 1.1 billion grasp poses across diverse real-world cluttered scenes.

File: Literature/graspnet_1912.13470v2.pdf [Full Academic PDF]

[070]

GraspView: Multi-View Spatial Awareness for Robust Robotic Manipulation

Grasping & Dexterity

Authors: Multi-View Perception Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2511.04199 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Integrates multi-view camera streams into a fused 3D voxel affordance grid to ensure occlusion-resilient grasp synthesis.

File: Literature/graspview_2511.04199v1.pdf [Full Academic PDF]

[071]

GraspVLA: Unified Visual-Grasp-Language-Action Representation for Mobile Manipulation

VLA & Foundation

Authors: VLA Manipulation Lab

Year: 2025 • Venue / Source: arXiv preprint arXiv:2505.03233 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Bridges high-level language task planning with low-level 6-DOF grasp pose generation within a unified transformer policy.

File: Literature/graspvla_2505.03233v3.pdf [Full Academic PDF]

[072]

HiBernac: Hierarchical Behavior Tree Synthesis from Natural Language

LLMs & Planning

Authors: Behavior Trees Research Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2506.08296 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Automatically synthesizes verifiable Hierarchical Behavior Trees (HBTs) from natural language instructions for safe robot execution.

File: Literature/hibernac_2506.08296v1.pdf [Full Academic PDF]

[073] Hierarchical LLM-Based Task Allocation and Motion Synthesis for Multi-Robot Systems

LLMs & Planning

Authors: Multi-Robot Systems Group

Year: 2026 • Venue / Source: arXiv preprint arXiv:2602.21670 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Employs hierarchical LLMs to decompose global collaborative assignments and synthesize decentralized trajectory plans across multiple arms.

File: Literature/hierarchical_llm_2602.21670v2.pdf

[Full Academic PDF]

[074] HiRobot: High-Precision Robotic Manipulation with Multi-Modal Foundation Policies

VLA & Foundation

Authors: Precision Manipulation Lab

Year: 2025 • Venue / Source: arXiv preprint arXiv:2502.19417 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Integrates high-resolution wrist camera features and fine force feedback to perform sub-millimeter precision assembly and manipulation.

File: Literature/hirobot_2502.19417v2.pdf

[Full Academic PDF]

[075] Confidence-Aware Human-in-the-Loop Robot Manipulation Policy

Teleop & Imitation

Authors: Human-Robot Collaboration Lab

Year: 2026 • Venue / Source: arXiv preprint arXiv:2602.10289 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Monitors epistemic policy uncertainty in real time, triggering proactive teleoperation intervention when confidence falls below safety thresholds.

File: Literature/hitl_confidence_aware_2602.10289v2.pdf

[Full Academic PDF]

[076] HybriK: A Hybrid Analytical-Neural Inverse Kinematics Solution for 3D Mesh Recovery and Robotics

Kinematics & Planning

Authors: Jiefeng Li, Chao Xu, Zhicun Chen, Siyuan Bian, Cewu Lu

Year: 2021 • Venue / Source: CVPR 2021 / arXiv:2011.14672 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Combines analytical closed-form IK equations with deep neural estimation to preserve accurate kinematic branch lengths and joint bounds.

File: Literature/hybrik_2011.14672v4.pdf

[Full Academic PDF]

[077] Learning 3D Grasp Synthesis from Demonstration: ICML Final

Grasping & Dexterity

Authors: Robot Learning Consortium

Year: 2011 • Venue / Source: International Conference on Machine Learning (ICML 2011) • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Early pioneering work on probabilistic grasp models learned from human demonstration data.

File: Literature/icml2011_final.pdf

[Full Academic PDF]

[078] In-Hand Object Rotation via Dexterous Multi-Fingered Manipulation

Grasping & Dexterity

Authors: Dexterous Robotics Laboratory

Year: 2022 • Venue / Source: arXiv preprint arXiv:2210.04887 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Formulates contact switching and continuous fingertip force redistribution to achieve 360-degree in-hand rotation of diverse shapes.

File: Literature/in_hand_rotation_2210.04887v1.pdf

[Full Academic PDF]

[079] Informed Circular Fields for Safe Dynamic Obstacle Avoidance in Manipulators

Kinematics & Planning

Authors: Motion Planning and Control Lab

Year: 2023 • Venue / Source: arXiv preprint arXiv:2212.05815 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Introduces circular harmonic vector fields for provably collision-free reactive navigation around moving obstacles in the arm workspace.

File: Literature/informed_circular_fields_2212.05815v2.pdf

[Full Academic PDF]

[080] Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models (LATS)

LLMs & Planning

Authors: Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, Yu-Xiong Wang

Year: 2024 • Venue / Source: ICLR 2024 / arXiv:2310.04406 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Unifies Monte Carlo Tree Search with LLM deliberation, state reflection, and environmental rollouts to solve complex multi-step tasks.

File: Literature/language_agent_tree_2310.04406v3.pdf

[Full Academic PDF]

[081] LargeVLM: Scaling Vision-Language Models for Autonomous Robotics and Scene Understanding

VLA & Foundation

Authors: Large VLM Robotics Group

Year: 2025 • Venue / Source: arXiv preprint arXiv:2508.13073 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Investigates scaling laws in Vision-Language Models applied to long-horizon spatial reasoning, affordance prediction, and robot action output.

File: Literature/largevlm_2508.13073v2.pdf

[Full Academic PDF]

[082] Learning Fast Manipulation via High-Speed Visual Feedback and Policy Adaptation

Teleop & Imitation

Authors: Fast Manipulation Research Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2508.20457 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Develops rapid policy adaptation techniques running at 100 Hz visual feedback rates for dynamic batting, catching, and agile placement.

File: Literature/learning_fast_2508.20457v1.pdf

[Full Academic PDF]

[083] Learning on the Job: Self-Improving Robot Manipulation Policies in Unseen Environments

Teleop & Imitation

Authors: Autonomous Adaptation Group

Year: 2023 • Venue / Source: arXiv preprint arXiv:2211.08416 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Presents an online self-supervised policy improvement system that fine-tunes robot manipulation skills during test-time deployment.

File: Literature/learning_on_the_job_2211.08416v3.pdf

[Full Academic PDF]

[084] LeRobot: Open-Source State-of-the-Art Machine Learning for Real-World Robotics

Teleop & Imitation

Authors: Remi Cadene, Simon Alibert, Alexander Soare, Quentin Gallouedec, et al. (Hugging Face)

Year: 2026 • Venue / Source: arXiv preprint arXiv:2602.22818 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: The Hugging Face open-source robotics library democratizing imitation learning, diffusion policies, and real-world teleoperation benchmarks.

File: Literature/lerobot_2602.22818v1.pdf

[Full Academic PDF]

[085] LLM-Map: Large Language Models for Semantic Topological Mapping and Navigation

LLMs & Planning

Authors: Semantic Mapping Laboratory

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.17309 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Constructs hierarchical topological graphs linked to LLM semantic embeddings for open-ended environment navigation.

File: Literature/llm_map_2503.17309v1.pdf

[Full Academic PDF]

[086] Prognostics and Health Management of the Robotic Servo-Motor under Variable Operating Conditions

PHM & Diagnostics

Authors: Shaolong Ling, et al.

Year: 2023 • Venue / Source: Mathematics (MDPI) (DOI: 10.3390/math11020398) •

Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Develops health indicator extraction algorithms for industrial robot joint servo-motors subject to dynamic speed and load cycles.

File: Literature/mathematics-11-00398.pdf

[Full Academic PDF]

[087] MegaPose: 6D Pose Estimation of Novel Objects via Render & Compare

Pose & Tracking

Authors: Yann Labbé, Lucas Manuelli, Arsalan Mousavian, Stephen Tyree, Stan Birchfield, et al.

Year: 2022 • Venue / Source: CoRL 2022 / arXiv:2212.06870 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Accurately predicts 6D poses of novel objects unseen during training via multi-view render-and-compare neural refinement.

File: Literature/megapose_2212.06870v1.pdf

[Full Academic PDF]

[088] MemGPT: Towards LLMs as Operating Systems

LLMs & Planning

Authors: Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, Joseph E. Gonzalez

Year: 2023 • Venue / Source: arXiv preprint arXiv:2310.08560 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Introduces virtual context management inspired by OS memory hierarchies, enabling LLM agents to maintain perpetual long-term memory.

File: Literature/memgpt_2310.08560v2.pdf

[Full Academic PDF]

[089] MiDaS v3.1 – A Model Zoo for Robust Monocular Relative Depth Estimation

3D Vision & GS

Authors: Reiner Birkel, Diana Wofk, Matthias Müller (Intel Labs)

Year: 2023 • Venue / Source: arXiv preprint arXiv:2307.14460 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: A comprehensive foundation model zoo for zero-shot relative depth estimation across varied sensors, camera geometries, and scenes.

File: Literature/midas_2307.14460v1.pdf

[Full Academic PDF]

[090] MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations

Teleop & Imitation

Authors: Ajay Mandlekar, Soroush Nasiriany, Bowen Wen, Iretiayo Akinola, Yashraj Narang, Linxi Fan, Yuke Zhu, Dieter Fox

Year: 2023 • Venue / Source: CoRL 2023 / arXiv:2310.17596 (NVIDIA) • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Scales a small set of human teleoperated demonstrations into thousands of diverse robotic manipulation trajectories automatically.

File: Literature/mimicgen_2310.17596v1.pdf

[Full Academic PDF]

[091] Using Mobile AR for Rapid Feasibility Analysis for Deployment of Robotic Arms

Middleware & Sensors

Authors: Mobile AR Robotics Group

Year: 2025 • Venue / Source: IEEE Robotics and Automation Letters (RA-L) • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Utilizes handheld Augmented Reality (AR) to overlay reachable workspace envelopes and verify robotic arm mounting feasibility in situ.

File: Literature/mobile_ar_2503.14725v2.pdf

[Full Academic PDF]

[092] A Model-Agnostic Approach for Semantically Driven Disambiguation in Human-Robot Interaction

LLMs & Planning

Authors: Semantic HRI Group

Year: 2024 • Venue / Source: arXiv preprint arXiv:2409.17004 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Formulates semantic disambiguation queries to clarify ambiguous user instructions in cooperative human-robot manipulation.

File: Literature/model-agnostic_2409.17004v2.pdf

[Full Academic PDF]

[093] Multi-Agent Systems for Robotic Autonomy with LLMs

LLMs & Planning

Authors: Junhong Chen, Ziqi Yang, Haoyuan G. Xu, Dandan Zhang, George Mylonas

Year: 2025 • Venue / Source: arXiv preprint arXiv:2505.05762 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Architects decentralized multi-agent LLM systems coordinating heterogeneous robot fleets in collaborative search and assembly.

File: Literature/multi-agent-2505.05762v1.pdf

[Full Academic PDF]

[094] Planning Multi-Fingered Grasps as Probabilistic Inference in a Learned Deep Network

Grasping & Dexterity

Authors: Deep Grasping Research Group

Year: 2018 • Venue / Source: arXiv preprint arXiv:1804.03289 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Casts multi-fingered grasp planning as maximum a posteriori inference over learned energy functions on 3D point clouds.

File: Literature/multi_fingered_grasps_1804.03289v1.pdf

[Full Academic PDF]

[095] Neural-Learning Trajectory Tracking Control of Flexible-Joint Robot Manipulators with Unknown Dynamics

Kinematics & Planning

Authors: Flexible Robotics Control Group

Year: 2019 • Venue / Source: arXiv preprint arXiv:1908.03269 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Designs adaptive neural network tracking controllers compensating for joint elasticity, nonlinear friction, and unmodeled arm dynamics.

File: Literature/neural_learning_1908.03269v1.pdf

[Full Academic PDF]

[096] Observe Then Act: Asynchronous Active Vision-Action Model for Robotic Manipulation

VLA & Foundation

Authors: Active Vision-Action Group

Year: 2025 • Venue / Source: IEEE Robotics and Automation Letters (RA-L) • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Decouples observation gathering from action execution to avoid motion blur and optimize camera gaze before arm motion.

File: Literature/observethenact_2409.14891v3.pdf

[Full Academic PDF]

[097] Octopus: On-Device Language Model for Function Calling of Software APIs

LLMs & Planning

Authors: Wei Chen, Zhiyuan Li, et al.

Year: 2024 • Venue / Source: arXiv preprint arXiv:2404.01549 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Designs ultra-compact on-device language models specialized in low-latency function calling and robotics software API execution.

File: Literature/octopus_2404.01549v2.pdf

[Full Academic PDF]

[098] Novel Demonstration Generation with Gaussian Splatting Enables Robust One-Shot Manipulation

3D Vision & GS

Authors: Gaussian Manipulation Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2504.13175 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Synthesizes thousands of synthetic camera viewpoints via 3D Gaussian Splatting from a single real demonstration, enabling robust one-shot learning.

File: Literature/one-shot-manip_2504.13175v1.pdf

[Full Academic PDF]

[099] Open-TeleVision: Teleoperation with Immersive Active Visual Feedback

Teleop & Imitation

Authors: Xuxin Cheng, Jialong Li, et al. (UC San Diego)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2407.01512 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Presents an open-source immersive teleoperation system using active stereo VR streaming for high-dexterity bimanual robot control.

File: Literature/open-television_2407.01512v2.pdf

[Full Academic PDF]

[100] OpenTeach: A Versatile Teleoperation System for Robotic Manipulation

Teleop & Imitation

Authors: Aadithya Iyer, Zhuoran Peng, Yinlong Dai, et al. (NYU)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2403.07870 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Provides a versatile teleoperation platform supporting smartphones, VR headsets, and hand tracking gloves across multiple robot embodiments.

File: Literature/openteach_2403.07870v1.pdf

[Full Academic PDF]

[101] OpenVLA: An Open-Source Vision-Language-Action Model

VLA & Foundation

Authors: Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, Chelsea Finn

Year: 2024 • Venue / Source: arXiv preprint arXiv:2406.09246 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: A 7B parameter open-source VLA model pretrained on 970k robot trajectories, achieving state-of-the-art zero-shot manipulation performance.

File: Literature/openvla_2406.09246v3.pdf

[Full Academic PDF]

[102] π 0.5: A Vision-Language-Action Model with Open-World Generalization

VLA & Foundation

Authors: Physical Intelligence Team (Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Chelsea Finn, et al.)

Year: 2025 • Venue / Source: arXiv preprint arXiv:2504.16054 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Physical Intelligence's generalist robot model unifying vision-language pretraining and flow-matching action chunking for dexterous tasks.

File: Literature/pi0.5_2504.16054v1.pdf

[Full Academic PDF]

[103] Prognostics and Health Management of the Robotic Servo-Motor under Variable Operating Conditions

PHM & Diagnostics

Authors: Shaolong Ling, et al.

Year: 2023 • Venue / Source: Mathematics (MDPI) • Predictive Maintenance, System Reliability & Diagnostics

Abstract / Core Focus: Models degradation and predicts failure onset in robot joint motors operating under dynamic velocity and torque profiles.

File: Literature/prognostics_mathematics-11-00398.pdf

[Full Academic PDF]

[104] Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

LLMs & Planning

Authors: Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V. Le, Denny Zhou (Google Brain)

Year: 2022 • Venue / Source: NeurIPS 2022 / arXiv:2201.11903 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: The foundational paper introducing Chain-of-Thought (CoT) prompting, drastically boosting multi-step symbolic and physical reasoning.

File: Literature/prompting_2201.11903v6.pdf

[Full Academic PDF]

[105] Reachability and Capability Analysis for Manipulation Tasks

Kinematics & Planning

Authors: Oliver Porges, Theodoros Stouraitis, Christoph Borst, et al. (DLR)

Year: 2020 • Venue / Source: German Aerospace Center (DLR) Robotics • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Analyzes 6D workspace reachability maps and manipulability ellipsoids for optimal base placement and grasp selection.

File: Literature/reachability.pdf

[Full Academic PDF]

[106] Reasoning with Language Model is Planning with World Model (RAP)

LLMs & Planning

Authors: Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, Zhiting Hu

Year: 2023 • Venue / Source: EMNLP 2023 / arXiv:2305.14992 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Proposes the RAP framework: repurposes LLMs as both world model simulators and heuristic search planners via MCTS.

File: Literature/reasoning_2305.14992v2.pdf

[Full Academic PDF]

[107] ReMEmbR: Building and Reasoning Over Long-Horizon Spatio-Temporal Memory for Robot Navigation

LLMs & Planning

Authors: Spatial Memory Group

Year: 2024 • Venue / Source: arXiv preprint arXiv:2409.13682 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Maintains persistent 3D spatio-temporal scene memory indexed by multimodal embeddings for long-horizon task execution.

File: Literature/remembr_2409.13682v1.pdf

[Full Academic PDF]

[108] RichMap: A Reachability Map Balancing Precision, Efficiency, and Flexibility for Rich Robot Manipulation Tasks

Kinematics & Planning

Authors: Manipulability Analysis Group

Year: 2026 • Venue / Source: arXiv preprint arXiv:2604.06778 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Introduces RichMap, a multi-resolution reachability representation that accelerates 6-DOF inverse kinematics lookup by orders of magnitude.

File: Literature/richmap_2604.06778v1.pdf

[Full Academic PDF]

[109]

Dedicated Scheduler for Kubernetes in Robotics Applications with Low Latency Requirements

Middleware & Sensors

Authors: André Santos, FEUP Robotics Lab
Year: 2023 • Venue / Source: Faculdade de Engenharia da Universidade do Porto (FEUP) •

Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Designs a deterministic low-latency Kubernetes pod scheduler optimized for cloud-robotics ROS 2 nodes and streaming control.

File: Literature/rigel_786600.pdf [Full Academic PDF]

[110]

RoboBrain: A Unified Brain Model for Robotic Manipulation from Abstract to Concrete

VLA & Foundation

Authors: Yuheng Ji, Huajie Tan, et al.
Year: 2025 • Venue / Source: arXiv preprint arXiv:2502.21257 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Proposes RoboBrain, integrating abstract symbolic planning with low-level continuous visual-motor skills into a single architecture.

File: Literature/robobrain_2502.21257v2.pdf [Full Academic PDF]

[111]

Spatial RoboGrasp: Generalized Robotic Grasping Control Policy

Grasping & Dexterity

Authors: Yiqi Huang, et al.
Year: 2025 • Venue / Source: arXiv preprint arXiv:2505.20814 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Learns spatial invariant representations from 3D point clouds to synthesize stable multi-angle parallel jaw grasps.

File: Literature/robograsp_2505.20814v1.pdf [Full Academic PDF]

[112]

Robo-GS: A Physics-Consistent Spatial-Temporal Model for Robotic Arms with Hybrid Representation

3D Vision & GS

Authors: Robo-GS Research Team
Year: 2024 • Venue / Source: arXiv preprint arXiv:2408.14873 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Couples 3D Gaussian Splatting with rigid-body kinematic priors to synthesize photorealistic digital twins of moving robotic arms in real time.

File: Literature/robogs_2408.14873v2.pdf [Full Academic PDF]

[113]

Robotic Perception of Transparent Objects: A Review

Pose & Tracking

Authors: Jiaqi Jiang, et al.
Year: 2023 • Venue / Source: IEEE Transactions on Artificial Intelligence (DOI: 10.1109/TAI.2023.3268800) •

Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Comprehensive survey on sensory challenges, polarization vision, neural radiance fields, and grasping algorithms for transparent glassware.

File: Literature/robotic_perception_2304.00157v2.pdf [Full Academic PDF]

[114]

Robowflex: Robot Motion Planning with MoveIt Made Easy

Kinematics & Planning

Authors: Zachary Kingston, Konstantinos Chamzas, Lydia E. Kavraki (Rice University)
Year: 2022 • Venue / Source: arXiv preprint arXiv:2103.12826 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Introduces Robowflex, an intuitive and flexible C++ library simplifying MoveIt motion planning, benchmarking, and scene configuration.

File: Literature/robowflex_2103.12826v2.pdf [Full Academic PDF]

[115] SAM-6D: Segment Anything Model Meets Zero-Shot 6D Object Pose Estimation

Pose & Tracking

Authors: Jiehong Lin, Lihua Liu, Dekun Lu, Kui Jia

Year: 2024 • Venue / Source: CVPR 2024 / arXiv:2311.15707 • Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Combines SAM segmentation with neural matching to solve zero-shot 6D object pose estimation from RGB-D inputs without target CAD models.

File: Literature/sam6d_2311.15707v2.pdf

[Full Academic PDF]

[116] Adaptive Grasp Pose Optimization for Robotic Arms Using Low-Cost RGB-D Sensors

Grasping & Dexterity

Authors: MDPI Sensors Robotics Authors

Year: 2025 • Venue / Source: Sensors (MDPI) (DOI: 10.3390/s25030909) • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Implements noise filtering and surface curvature analysis on low-cost RGB-D depth feeds to optimize grasp pose selection.

File: Literature/sensors-25-00909-with-cover.pdf

[Full Academic PDF]

[117] Application of Cloud Simulation Techniques for Robotic Software Validation

Middleware & Sensors

Authors: Diogo Vieira, Miguel Martins, et al.

Year: 2025 • Venue / Source: Sensors (MDPI) (DOI: 10.3390/s25051693) • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Presents a cloud-native CI/CD simulation testbed executing headless Gazebo and ROS 2 test suites for software regression validation.

File: Literature/sensors-25-01693.pdf

[Full Academic PDF]

[118] Time and Frequency Domain Analysis of IMU-Based Orientation Estimation Algorithms with Comparison to Robotic Manipulators

Middleware & Sensors

Authors: Sensor Fusion Research Team

Year: 2025 • Venue / Source: Sensors (MDPI) (DOI: 10.3390/s25165161) • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Benchmarks Madgwick, Mahony, and Extended Kalman Filters against high-accuracy robotic arm optical ground truth for joint angle estimation.

File: Literature/sensors-25-05161-v2.pdf

[Full Academic PDF]

[119] Deformable and Fragile Object Manipulation: A Review and Prospects

Grasping & Dexterity

Authors: Yicheng Zhu, et al.

Year: 2025 • Venue / Source: Sensors (MDPI) (DOI: 10.3390/s25175430) • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Surveys tactile sensing arrays, compliance control, and finite-element modeling for handling delicate agricultural and food produce.

File: Literature/sensors-25-05430-v2.pdf

[Full Academic PDF]

[120] Singularity Avoidance in Inverse Kinematics: A Unified Treatment of Classical and Learning-Based Methods

Kinematics & Planning

Authors: Inverse Kinematics Theory Group

Year: 2026 • Venue / Source: arXiv preprint arXiv:2604.13405 • Kinematics, Dynamics & Motion Planning

Abstract / Core Focus: Unifies damped least squares, manipulability gradient projection, and neural barrier certificates to prevent arm singularities.

File: Literature/singularity_avoidance_2604.13405v1.pdf

[Full Academic PDF]

[121] Optimizing Small Language Models for In-Vehicle Function-Calling

LLMs & Planning

Authors: Yahya Sowti, et al.

Year: 2025 • Venue / Source: arXiv preprint arXiv:2501.02342 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Quantizes and fine-tunes 1B-3B parameter SLMs for ultra-fast deterministic JSON function execution in resource-constrained hardware.

File: Literature/small_llm_2501.02342v1.pdf

[Full Academic PDF]

[122] SMART-LLM: Smart Multi-Agent Robot Task Planning using Large Language Models

LLMs & Planning

Authors: Yiding Chen, et al.

Year: 2023 • Venue / Source: arXiv preprint arXiv:2309.10062 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Decomposes complex human instructions into multi-agent subtasks and optimizes task allocation across diverse robots.

File: Literature/smart_llm_2309.10062v2.pdf

[Full Academic PDF]

[123] Learning Manipulation Skills through Robot Chain-of-Thought with Sparse Failure Guidance

VLA & Foundation

Authors: Robot CoT Research Team

Year: 2024 • Venue / Source: arXiv preprint arXiv:2405.13573 • Vision-Language-Action (VLA) & Foundation Models

Abstract / Core Focus: Utilizes sparse failure labels during policy execution to refine intermediate reasoning tokens in embodied chain-of-thought models.

File: Literature/sparse_guidance_2405.13573v3.pdf

[Full Academic PDF]

[124] 3D Gaussian Splatting in Robotics: A Survey

3D Vision & GS

Authors: Siting Zhu, Guangming Wang, Xinjie Wang, Hesheng Wang (Shanghai Jiao Tong University)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2410.12262 • 3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: First comprehensive survey on 3D Gaussian Splatting applications in robotics: SLAM, dynamic tracking, simulation, and manipulation.

File: Literature/splatting_2410.12262v2.pdf

[Full Academic PDF]

[125] SpringGrasp: Synthesizing Compliant, Dexterous Grasps under Shape Uncertainty

Grasping & Dexterity

Authors: Sirui Chen, Jeannette Bohg, C. Karen Liu (Stanford University)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2404.13532 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Models virtual compliance springs at finger contact points to guarantee stable dexterous grasps under severe object shape uncertainty.

File: Literature/springgrasp_2404.13532v1.pdf

[Full Academic PDF]

[126] Tactile-Conditioned Diffusion Policy for Force-Aware Robotic Manipulation

Grasping & Dexterity

Authors: Tactile Robotics Lab

Year: 2025 • Venue / Source: arXiv preprint arXiv:2510.13324 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Conditions diffusion action policies on continuous high-density tactile sensor arrays for delicate insertion and contact-rich assembly.

File: Literature/tactile-condition_2510.13324v1.pdf

[Full Academic PDF]

[127]

Talk-to-Resolve: Combining Scene Understanding and Spatial Dialogue to Resolve Ambiguities

LLMs & Planning

Authors: Interactive Robotics Lab

Year: 2022 • Venue / Source: arXiv preprint arXiv:2111.11099 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Couples 3D object detection with multimodal dialogue generation to resolve referring expressions in complex tabletop scenes.

File: Literature/talk_to_resolve_2111.11099v2.pdf

[Full Academic PDF]

[128]

Improving Reproducibility in Machine Learning: Overview, Barriers, and Drivers

Middleware & Sensors

Authors: Thulani Myles Mzyece, et al.

Year: 2026 • Venue / Source: TechRxiv Preprint • Robotics Frameworks, Calibration & Sensor Interfaces

Abstract / Core Focus: Analyzes reproducibility barriers across deep learning pipelines and presents protocols for deterministic robotic model validation.

File: Literature/techrxiv.177155769.94899097_v1.pdf

[Full Academic PDF]

[129]

ThinkGrasp: A Vision-Language System for Strategic Part Grasping in Clutter

Grasping & Dexterity

Authors: Yaoyao Qian, Xupeng Zhu, Ondrej Biza, et al.

Year: 2024 • Venue / Source: arXiv preprint arXiv:2407.11298 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Combines VLM spatial reasoning with geometric grasping networks to strategize part-level grasping in heavily stacked clutter.

File: Literature/thinkgrasp_2407.11298v2.pdf

[Full Academic PDF]

[130]

TinyAgent: Function Calling at the Edge

LLMs & Planning

Authors: Lutfi Eren Erdogan, Nicholas Lee, Siddharth Jha, Sehoon Kim, Ryan Tabrizi, et al. (UC Berkeley)

Year: 2024 • Venue / Source: arXiv preprint arXiv:2409.00608 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Demonstrates high-accuracy function calling on tiny 1B-3B parameter edge models outperforming larger models via specialized tool tuning.

File: Literature/tinyagent_2409.00608v3.pdf

[Full Academic PDF]

[131]

TransCG: A Large-Scale Real-World Dataset for Transparent Object Grasping and Depth Estimation

3D Vision & GS

Authors: Hongjie Fang, Hao-Shu Fang, Sheng Xu, Cewu Lu (Shanghai Jiao Tong University)

Year: 2022 • Venue / Source: IEEE Robotics and Automation Letters (RA-L) / arXiv:2202.08471 •

3D Vision, Gaussian Splatting & Spatial Representations

Abstract / Core Focus: Releases a comprehensive real-world transparent object dataset featuring 51,000 paired RGB-D images and 6-DOF grasp ground truth.

File: Literature/transcg_2202.08471v2.pdf

[Full Academic PDF]

[132]

Tree of Thoughts: Deliberate Problem Solving with Large Language Models

LLMs & Planning

Authors: Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, Karthik Narasimhan (Princeton & Google DeepMind)

Year: 2023 • Venue / Source: NeurIPS 2023 / arXiv:2305.10601 • LLM Task Planning, Multi-Agent Reasoning & Decision Making

Abstract / Core Focus: Introduces Tree of Thoughts (ToT), enabling language models to explore multiple reasoning paths, evaluate choices, and backtrack.

File: Literature/tree_of_thoughts_2305.10601v2.pdf

[Full Academic PDF]

[133] Uncertainty Comes for Free: Human-in-the-Loop Policies with Diffusion Models

Teleop & Imitation

Authors: Diffusion Robotics Team

Year: 2025 • Venue / Source: arXiv preprint arXiv:2503.01876 • Teleoperation, Demonstration & Robot Learning

Abstract / Core Focus: Extracts ensemble variance and score entropy directly from diffusion policy denoising trajectories to trigger timely human intervention.

File: Literature/uncertainty_comes_for_free_2503.01876v3.pdf [Full Academic PDF]

[134] Towards a Unified Understanding of Robot Manipulation: A Comprehensive Survey

Grasping & Dexterity

Authors: Unified Manipulation Survey Consortium

Year: 2025 • Venue / Source: arXiv preprint arXiv:2510.10903 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Presents a unified taxonomy linking perception, affordances, trajectory generation, tactile feedback, and foundation policy learning.

File: Literature/unified_robot_manipulation_2510.10903v1.pdf [Full Academic PDF]

[135] Visual Dexterity: In-Hand Reorientation of Novel and Complex Object Shapes

Grasping & Dexterity

Authors: Tao Chen, Megha Tippur, Siyang Qin, Edward Adelson, Pulkit Agrawal (MIT)

Year: 2023 • Venue / Source: Science Robotics / arXiv:2211.11744 • Robotic Grasping & Dexterous Manipulation

Abstract / Core Focus: Demonstrates arbitrary in-hand reorientation of complex geometric objects using vision and proprioception on an anthropomorphic hand.

File: Literature/visual_dexterity_2211.11744v3.pdf [Full Academic PDF]

[136] Wide-Range and Real-Time 3D Tracking of a Moving Object by Dynamic Vision Sensor

Pose & Tracking

Authors: Journal of Robotics and Mechatronics Authors

Year: 2025 • Venue / Source: Journal of Robotics and Mechatronics (DOI: 10.20965/jrm.2025.p0399) •

Object Pose Estimation, Tracking & Perception

Abstract / Core Focus: Employs neuromorphic Dynamic Vision Sensors (event cameras) for ultra-low latency wide-range 3D tracking of high-speed moving targets.

File: Literature/wide_range_tracking_Fujipress_JRM-37-2-13.pdf [Full Academic PDF]

Thematic Cross-Reference Index by Research Domain

9 Categories • 136 Indexed Papers

• Teleoperation, Demonstration & Robot Learning (10 Papers)

[001] Recent Advances in Robot Learning from Demonstration (2020) 2019_LfD_Survey_preprint.pdf

[014] GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators (2024) GELLO_2309.13037v2.pdf

[075] Confidence-Aware Human-in-the-Loop Robot Manipulation Policy (2026) hitl_confidence_aware_2602.10289v2.pdf

[082] Learning Fast Manipulation via High-Speed Visual Feedback and Policy Adaptation (2025) learning_fast_2508.20457v1.pdf

[083] Learning on the Job: Self-Improving Robot Manipulation Policies in Unseen Environments (2023) learning_on_the_job_2211.08416v3.pdf

[084] LeRobot: Open-Source State-of-the-Art Machine Learning for Real-World Robotics (2026) lerobot_2602.22818v1.pdf

[090] MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations (2023) mimicgen_2310.17596v1.pdf

[099] Open-TeleVision: Teleoperation with Immersive Active Visual Feedback (2024) open-television_2407.01512v2.pdf

[100] OpenTeach: A Versatile Teleoperation System for Robotic Manipulation (2024) openteach_2403.07870v1.pdf

[133] Uncertainty Comes for Free: Human-in-the-Loop Policies with Diffusion Models (2025) uncertainty_comes_for_free_2503.01876v3.pdf

• 3D Vision, Gaussian Splatting & Spatial Representations (14 Papers)

[002] 3D Gaussian Splatting for Real-Time Radiance Field Rendering (2023) 3d_gaussian_2308.04079v1.pdf

[003] Foundations of Spatial Perception for Robotics: Hierarchical Representations and Real-time Systems (2023) 3d_rep_2305.07154v1.pdf

[004] 3D VSG: Long-term Semantic Scene Change Prediction through 3D Variable Scene Graphs (2023) 3dvsg_2209.07896v2.pdf

[009] Assessing Depth Anything V2 Monocular Depth Estimation as a LiDAR Alternative in Robotics (2025) Assessing_Depth_Anything_V2_monocular_depth_estima.pdf

[023] SAM 2: Segment Anything in Images and Videos (2024) SAM2_2408.00714v2.pdf

[046] ClearDepth: High-Fidelity Depth Completion for Transparent Objects (2024) cleardepth_2409.08926v3.pdf

[047] ClearGrasp: 3D Shape Estimation and Grasping of Transparent Objects (2020) cleargrasp_1910.02550v2.pdf

[052] Depth Anything V2: A Deep Monocular Depth Estimation Foundation Model (2024) depthanything_v2_2406.09414v2.pdf

[068] Graph-EQA: Embodied Question Answering on 3D Dynamic Scene Graphs (2023) graph_EQA_2305.07154v1.pdf

[089] MiDaS v3.1 – A Model Zoo for Robust Monocular Relative Depth Estimation (2023) midas_2307.14460v1.pdf

[098] Novel Demonstration Generation with Gaussian Splatting Enables Robust One-Shot Manipulation (2025) one-shot-manip_2504.13175v1.pdf

[112] Robo-GS: A Physics-Consistent Spatial-Temporal Model for Robotic Arms with Hybrid Representation (2024) robogs_2408.14873v2.pdf

[124] 3D Gaussian Splatting in Robotics: A Survey (2024) splatting_2410.12262v2.pdf

[131] TransCG: A Large-Scale Real-World Dataset for Transparent Object Grasping and Depth Estimation (2022) transcg_2202.08471v2.pdf

• Kinematics, Dynamics & Motion Planning

(19 Papers)

- [005] AI-Enhanced Kinematic Modeling of Flexible Manipulators Using Multi-IMU Sensor Fusion (2025) [AI_enhanced_kinematic_modelling_2510.02975v1.pdf](#)
- [008] A Multi-Robot Collaborative Manipulation Framework with Real-Time Distributed Optimization (2025) [A_multi-robot_collaborative_manipulation_framework.pdf](#)
- [016] Manipulator Robot Development and Analysis of Motion Planning Libraries Through ROS2 and MoveIt2 (2023) [ManipulatorRobotDevelopment.pdf](#)
- [018] Neural Inverse Kinematics (2022) [Neural_IK_2205.10837v1.pdf](#)
- [028] Deep Learning Approaches for Robotic Arm Trajectory Prediction and Kinematics (2024) [IJISAE_2_KAVITHA+VIJAYANAND_6_3061.pdf](#)
- [035] Avoid Everything: Real-Time Dynamic Obstacle Avoidance for Robot Manipulators (2025) [avoid_everything_fishman25a.pdf](#)
- [040] Analytical and Deep Learning Approaches for Solving the Inverse Kinematic Problem of High-DOF Robotic Arms (2023) [citation-372419852.txt](#)
- [042] A Novel Graph Neural Network Approach for Inverse Kinematics in Robotic Arms (2025) [citation-392564836.txt](#)
- [044] High-Stiffness Path Planning for 7-DOF Cable-Driven Manipulators in Single and Dual-Arm Configurations (2025) [citation-398061715.txt](#)
- [048] Closed-Loop Visuomotor Control for Precise Robotic Manipulation (2022) [closed_loop_2207.10543v1.pdf](#)
- [049] Closed-Loop Kinematic Chains for Real-Time Robot Calibration (2017) [closed_loop_chain_1612.01650v3.pdf](#)
- [050] cuRobo: Parallelized GPU-Accelerated Robot Motion Generation and Collision Checking (2024) [curobo_2310.17274v2.pdf](#)
- [076] HybriK: A Hybrid Analytical-Neural Inverse Kinematics Solution for 3D Mesh Recovery and Robotics (2021) [hybrik_2011.14672v4.pdf](#)
- [079] Informed Circular Fields for Safe Dynamic Obstacle Avoidance in Manipulators (2023) [informed_circular_fields_2212.05815v2.pdf](#)
- [095] Neural-Learning Trajectory Tracking Control of Flexible-Joint Robot Manipulators with Unknown Dynamics (2019) [neural_learning_1908.03269v1.pdf](#)
- [105] Reachability and Capability Analysis for Manipulation Tasks (2020) [reachability.pdf](#)
- [108] RichMap: A Reachability Map Balancing Precision, Efficiency, and Flexibility for Rich Robot Manipulation Tasks (2026) [richmap_2604.06778v1.pdf](#)
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- [120] Singularity Avoidance in Inverse Kinematics: A Unified Treatment of Classical and Learning-Based Methods (2026) [singularity_avoidance_2604.13405v1.pdf](#)

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- [006] ARC-Calib: Autonomous Markerless Camera-to-Robot Calibration via Exploratory Robot Motions (2025) [ARC_Calib_2503.14701v1.pdf](#)
- [007] A ROS-Gazebo Interface for the Katana Robotic Arm Manipulation (2021) [A_ROS-Gazebo_Interface_for_the_Katana_Robotic_Arm_.pdf](#)
- [017] MoveIt! and ROS Topics Architecture for Robotic Manipulation (2022) [MoveIt_ROS_Topics.pdf](#)
- [022] Robot Operating System 2: Design, Architecture, and Uses In The Wild (2022) [ROS2_2211.07752v1.pdf](#)
- [036] AprilTag: A Robust and Flexible Visual Fiducial System (2011) [citation-224252184.txt](#)
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- [038] Improved State Estimation of a Robot End-Effector Using Laser Tracker and Inertial Sensor Fusion (2022) [citation-360052156.txt](#)
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- [091] Using Mobile AR for Rapid Feasibility Analysis for Deployment of Robotic Arms (2025) [mobile_ar_2503.14725v2.pdf](#)
- [109] Dedicated Scheduler for Kubernetes in Robotics Applications with Low Latency Requirements (2023) [rigel_786600.pdf](#)
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[010] Autonomous Object Tracking with Vision-Based Control Using a 2-DOF Robotic Arm (2025)	Autonomous_object_tracking_with_vision_based_contr.pdf
[013] EXOT: Exit-Aware Object Tracker for Safe Robotic Manipulation of Moving Objects (2023)	EXOT_2306.05262v1.pdf
[024] Speed-FairMOT: Multi-Class Multi-Object Tracking for Real-Time Robotic Applications (2023)	Speed-FairMOT_multi-class_multi-object_tracking_fo.pdf
[029] Active Perception for Robotic Manipulation: A Survey (2023)	active_perception_2310.00433v2.pdf
[030] ActivePose: Active 6D Object Pose Estimation via Viewpoint Planning (2025)	activepose_2509.11364v2.pdf
[034] Any6D: Zero-Shot Category-Level 6D Object Pose Estimation (2025)	any6d_2503.18673v2.pdf
[055] Dynamic Open-Vocabulary Object Perception for Mobile Manipulation (2024)	dynamic_openvocab_2410.11989v6.pdf
[059] FastTracker: Real-Time High-Speed 3D Object Tracking for Dynamic Robotic Grasping (2025)	fasttracker_2508.14370v1.pdf
[062] FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects (2024)	foundationpose_2312.08344v2.pdf
[064] GigaPose: Fast and Robust Novel Object Pose Estimation via CAD Models (2024)	gigapose_2311.14155v2.pdf
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• Vision-Language-Action (VLA) & Foundation Models

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[011] Robotic Control via Embodied Chain-of-Thought Reasoning (2025)	COT_2407.08693v3.pdf
[012] CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models (2025)	COT_VLA_2503.22020v1.pdf
[019] Octo: An Open-Source Generalist Robot Policy (2024)	OCTO_2405.12213v2.pdf
[026] A Survey on Reinforcement Learning of Vision-Language-Action Models for Robotic Manipulation (2025)	TechrxivA_Survey_on_Reinforcement_Learning_of_Vision-Language-Action_Models_for_Robotic_Manipulation.pdf
[027] Vision-Language-Action Models in Robotic Manipulation: A Comprehensive Review (2024)	Vision_Language_Action_Models_in_Robotic_Manipulat.pdf
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[054] Do As I Can, Not As I Say: Grounding Language in Robotic Affordances (SayCan) (2022)	doasican_2204.01691v2.pdf
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[065] GLOVER: Generalizable Language-Guided Robotic Policy via Visual Goal Verification (2024)	glover_2411.12286v2.pdf
[066] GR00T-N1: An Open-Source Foundation Model for Generalist Humanoid and Arm Manipulation (2025)	gr00t_n1_2503.14734v2.pdf
[067] GR-3 Technical Report: Next-Generation Generalist Robot Manipulation System (2025)	gr3_tech_report_2507.15493v2.pdf
[071] GraspVLA: Unified Visual-Grasp-Language-Action Representation for Mobile Manipulation (2025)	graspvla_2505.03233v3.pdf
[074] HiRobot: High-Precision Robotic Manipulation with Multi-Modal Foundation Policies (2025)	hirobot_2502.19417v2.pdf
[081] LargeVLM: Scaling Vision-Language Models for Autonomous Robotics and Scene Understanding (2025)	largevlm_2508.13073v2.pdf
[096] Observe Then Act: Asynchronous Active Vision-Action Model for Robotic Manipulation (2025)	observethenact_2409.14891v3.pdf
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• LLM Task Planning, Multi-Agent Reasoning & Decision Making

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[015] LLM-Based Ambiguity Detection in Natural Language Instructions for Collaborative Surgical Robots (2025)	LLM-Based_2507.11525v1.pdf
[039] Asking Follow-Up Clarifications to Resolve Ambiguities in Human-Robot Conversation (2022)	citation-364050842.txt
[051] Decision Making with Large Language Models for Autonomous Robotics (2023)	decision_making_2303.06710v2.pdf
[072] HiBernac: Hierarchical Behavior Tree Synthesis from Natural Language (2025)	hibernac_2506.08296v1.pdf
[073] Hierarchical LLM-Based Task Allocation and Motion Synthesis for Multi-Robot Systems (2026)	hierarchical_llm_2602.21670v2.pdf
[080] Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models (LATS) (2024)	language_agent_tree_2310.04406v3.pdf
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